ENGINEERS WALLAH

# CHEMISTRY

## Lecture III

## Let's Explore Valency, Elements and Ions

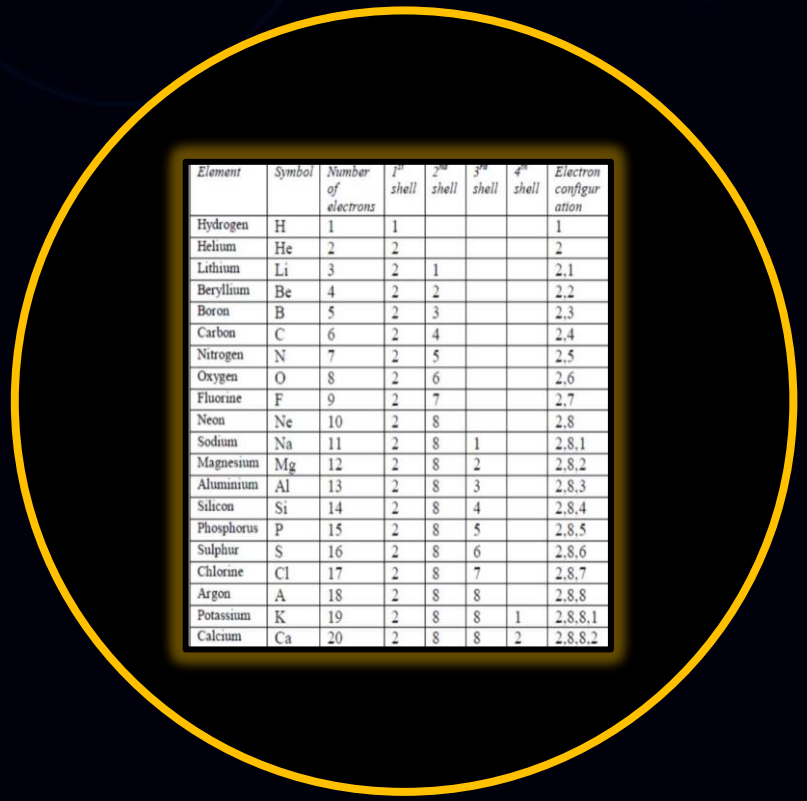



By: Sunil Bhaiya

# TOPICS *to be covered*

1. Valency and Its Calculation
2. Elements and Their Relation with Molecules
3. Ions and Its Types

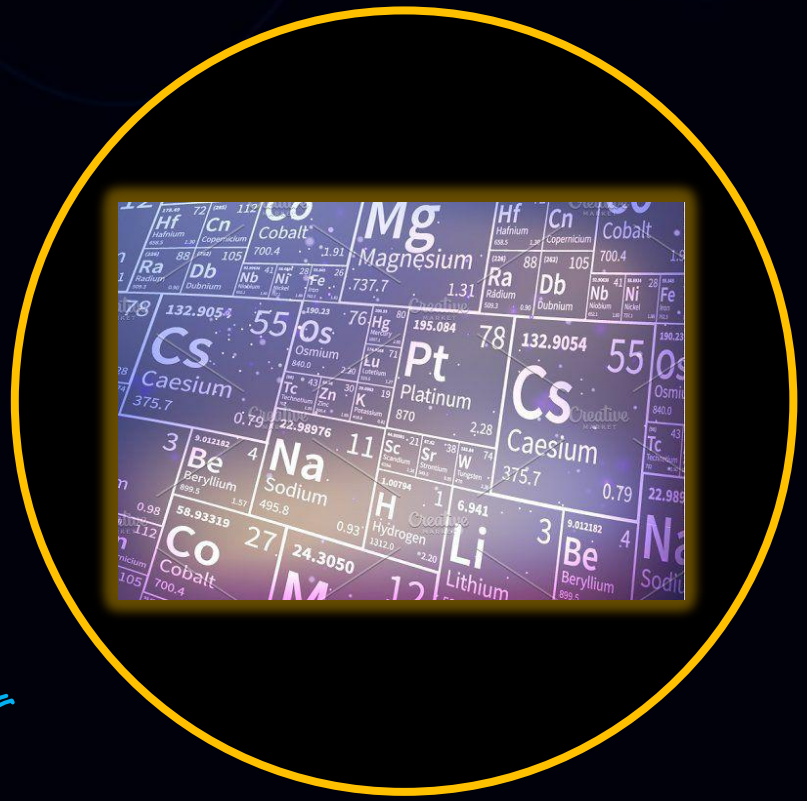
# Knowledge Ride On



| Element | Symbol | Number of electrons | 1st shell | 2nd shell | 3rd shell | 4th shell | Electron configuration |
|---|---|---|---|---|---|---|---|
| Hydrogen | H | 1 | 1 | | | | 1 |
| Helium | He | 2 | 2 | | | | 2 |
| Lithium | Li | 3 | 2 | 1 | | | 2.1 |
| Beryllium | Be | 4 | 2 | 2 | | | 2.2 |
| Boron | B | 5 | 2 | 3 | | | 2.3 |
| Carbon | C | 6 | 2 | 4 | | | 2.4 |
| Nitrogen | N | 7 | 2 | 5 | | | 2.5 |
| Oxygen | O | 8 | 2 | 6 | | | 2.6 |
| Fluorine | F | 9 | 2 | 7 | | | 2.7 |
| Neon | Ne | 10 | 2 | 8 | | | 2.8 |
| Sodium | Na | 11 | 2 | 8 | 1 | | 2.8.1 |
| Magnesium | Mg | 12 | 2 | 8 | 2 | | 2.8.2 |
| Aluminium | Al | 13 | 2 | 8 | 3 | | 2.8.3 |
| Silicon | Si | 14 | 2 | 8 | 4 | | 2.8.4 |
| Phosphorus | P | 15 | 2 | 8 | 5 | | 2.8.5 |
| Sulphur | S | 16 | 2 | 8 | 6 | | 2.8.6 |
| Chlorine | Cl | 17 | 2 | 8 | 7 | | 2.8.7 |
| Argon | A | 18 | 2 | 8 | 8 | | 2.8.8 |
| Potassium | K | 19 | 2 | 8 | 8 | 1 | 2.8.8.1 |
| Calcium | Ca | 20 | 2 | 8 | 8 | 2 | 2.8.8.2 |

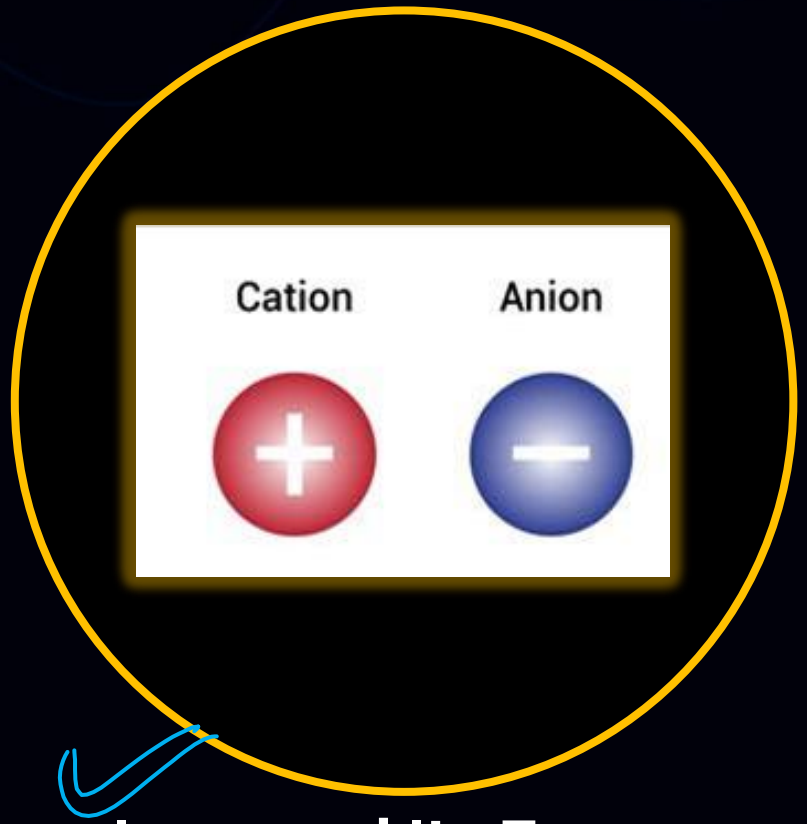
## Valency and Its Calculation

# Knowledge Ride On




## Elements and Their Relation with Molecules

# Knowledge Ride On




## Ions and Its Types

# Knowledge Ride On



Efficiency Hacks



# Glossary

Valency (संयोजकता)
Molecule (अणु)
Cation (धनायन)
Anion (ऋणायन)
Element (तत्व)
Noble gas (उत्कृष्ट गैस)

# Practice and PYQ's Wallah



PW Ka ChemStar!

# Question

When dry ice is heated, it changes directly into _____.

SSC JE 2020

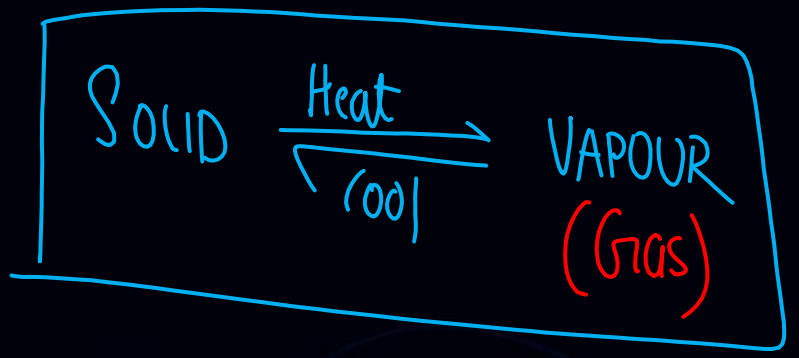
Solid carbon dioxide → Sublimable substance → It shows sublimation.

A gas

B powder

C solid

D liquid



## Question

Which among the following scientist discovered electron in 1897?

SSC JE 2017

A. Ernest Rutherford

B. Carl D. Anderson

C. J. J. Thomson (1897)

D. Joseph Priestly

## Question

Who discovered the neutron and its properties.

*SSC JE 2017*

A Nikola Tesla

B James Chadwick (1932)

C Albert Einstein

D Ernest Rutherford

**Question**

Which element has the least number of valence electrons?

RRB JE 2015

Outermost shell electrons

| | | A.No. | K | L | M | N |
|---|---|---|---|---|---|---|
| A | Chlorine | 17 | 2 | 8 | 7 | |
| B ✓ | Sodium | 11 | 2 | 8 | 1 | |
| C | Fluorine | 9 | 2 | 7 | | |
| D | Oxygen | 8 | 2 | 6 | | |

## Question

The maximum number of electrons that can be accommodated in M shell of an atom is _____.

RRB JE 2015

A 2

B 8

C 18

D 32

K, $n=1$, $2\times(1)^2 = 2$ electrons

L, $n=2$, $2\times(2)^2 = 8$ electrons

M, $n=3$, $2\times(3)^2 = 18$ electrons

N, $n=4$, $2\times(4)^2 = 32$ electrons

# Concept Bites – Valency and Its Calculation

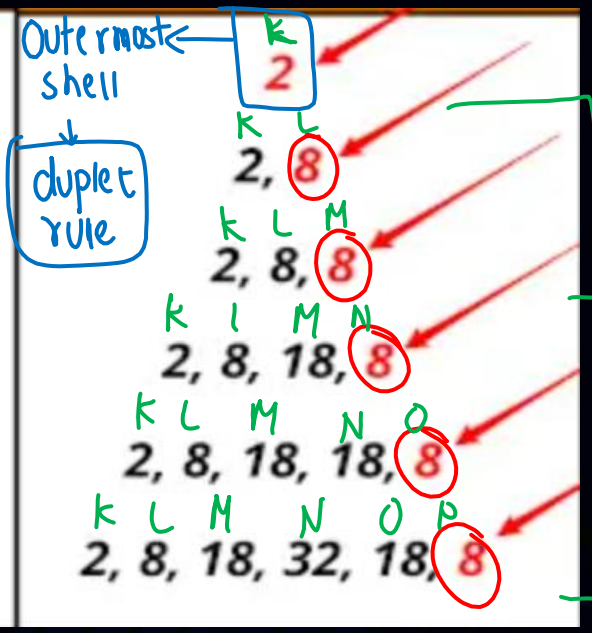
# Valency (संयोजकता)

how many electron(s) an atom will lose, gain or share.

The combining capacity of an atom of an element to attain nearest stable noble gas configuration.

C-I → *Kossel-Lewis approach (Octet Rule)* → Observed and analysed noble gases / inert gas → unreactive → Stable electronic configuration

| | | |
|---|---|---|
| Heena ← | Helium (He) | 2 |
| Neena ← | Neon (Ne) | 10 |
| Aur ← | Argon (Ar) | 18 |
| Kareena ← | Krypton (Kr) | 36 |
| X-Ray ← | Xenon (Xe) | 54 |
| Rangeen ← | Radon (Rn) | 86 |




outermost shell → 8 electrons

# Calculation of Valency

If outermost shell (valence shell) has 1, 2, 3, 4 electrons then the valency will be:

1, 2, 3, 4

If outermost shell has 5, 6 and 7 electrons then the valency will be:

8-5 = 3, 8-6 = 2, 8-7 = 1

Valency of noble gases is: 'ZERO'

0

| A·No.(z) | Element | Symbol | Number of electrons | 1st shell K | 2nd shell L | 3rd shell M | 4th shell N | Electron configuration | Valency |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Hydrogen | H | 1 | 1 | | | | 1 | 1 |
| 2 | Helium | He | 2 | 2 | | | | 2 | → 0 |
| 3 | Lithium | Li | 3 | 2 | 1 | | | 2.1 | 1 |
| 4 | Beryllium | Be | 4 | 2 | 2 | | | 2.2 | 2 |
| 5 | Boron | B | 5 | 2 | 3 | | | 2.3 | 3 |
| 6 | Carbon | C | 6 | 2 | 4 | | | 2.4 | 4 |
| 7 | Nitrogen | N | 7 | 2 | 5 | | | 2.5 | (8-5) → 3 |
| 8 | Oxygen | O | 8 | 2 | 6 | | | 2.6 | (8-6) → 2 |
| 9 | Fluorine | F | 9 | 2 | 7 | | | 2.7 | (8-7) → 1 |
| 10 | Neon | Ne | 10 | 2 | 8 | | | 2.8 | → 0 |
| 11 | Sodium | Na | 11 | 2 | 8 | 1 | | 2.8.1 | → 1 |
| 12 | Magnesium | Mg | 12 | 2 | 8 | 2 | | 2.8.2 | → 2 |
| 13 | Aluminium | Al | 13 | 2 | 8 | 3 | | 2.8.3 | → 3 |
| 14 | Silicon | Si | 14 | 2 | 8 | 4 | | 2.8.4 | → 4 |
| 15 | Phosphorus | P | 15 | 2 | 8 | 5 | | 2.8.5 | (8-5) → 3 |
| 16 | Sulphur | S | 16 | 2 | 8 | 6 | | 2.8.6 | (8-6) → 2 |
| 17 | Chlorine | Cl | 17 | 2 | 8 | 7 | | 2.8.7 | (8-7) → 1 |
| 18 | Argon | Ar | 18 | 2 | 8 | 8 | | 2.8.8 | → 0 |
| 19 | Potassium | K | 19 | 2 | 8 | 8 | 1 | 2.8.8.1 | → 1 |
| 20 | Calcium | Ca | 20 | 2 | 8 | 8 | 2 | 2.8.8.2 | → 2 |

# Practice and PYQ's Wallah



PW Ka ChemStar!

# Question

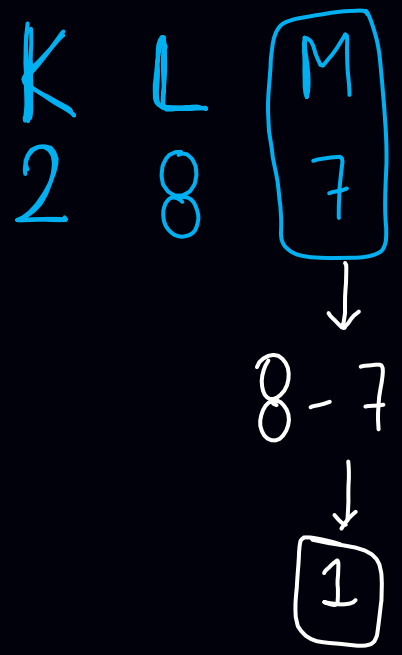
What is the valency of an element with 17 electrons?

A 1

B 7

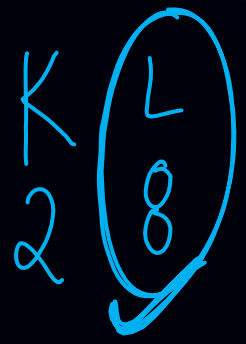
C 3

D 4



# Question

What is the valency of an element with 10 electrons?




A 1

B 0

C 2

D 3

**Question**

Which of the following is not a noble gas?

*RRB JE 2014*

A Helium

B Bromine

C Argon

D Neon

# Concept Bites – Element and Its Relation with Molecules

Coined by → Robert Boyle (1661)

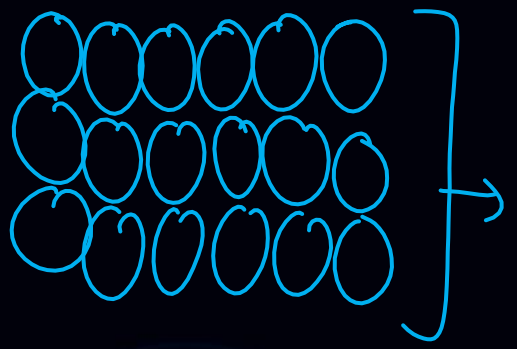
A pure substance is one that is made up of the same type of particles, i.e. it is chemically uniform or homogeneous.

chemically → joined together by force of attraction

uniform or homogeneous → एक समान

- Elements (तत्व)
- Compounds

Iron (लोहा)



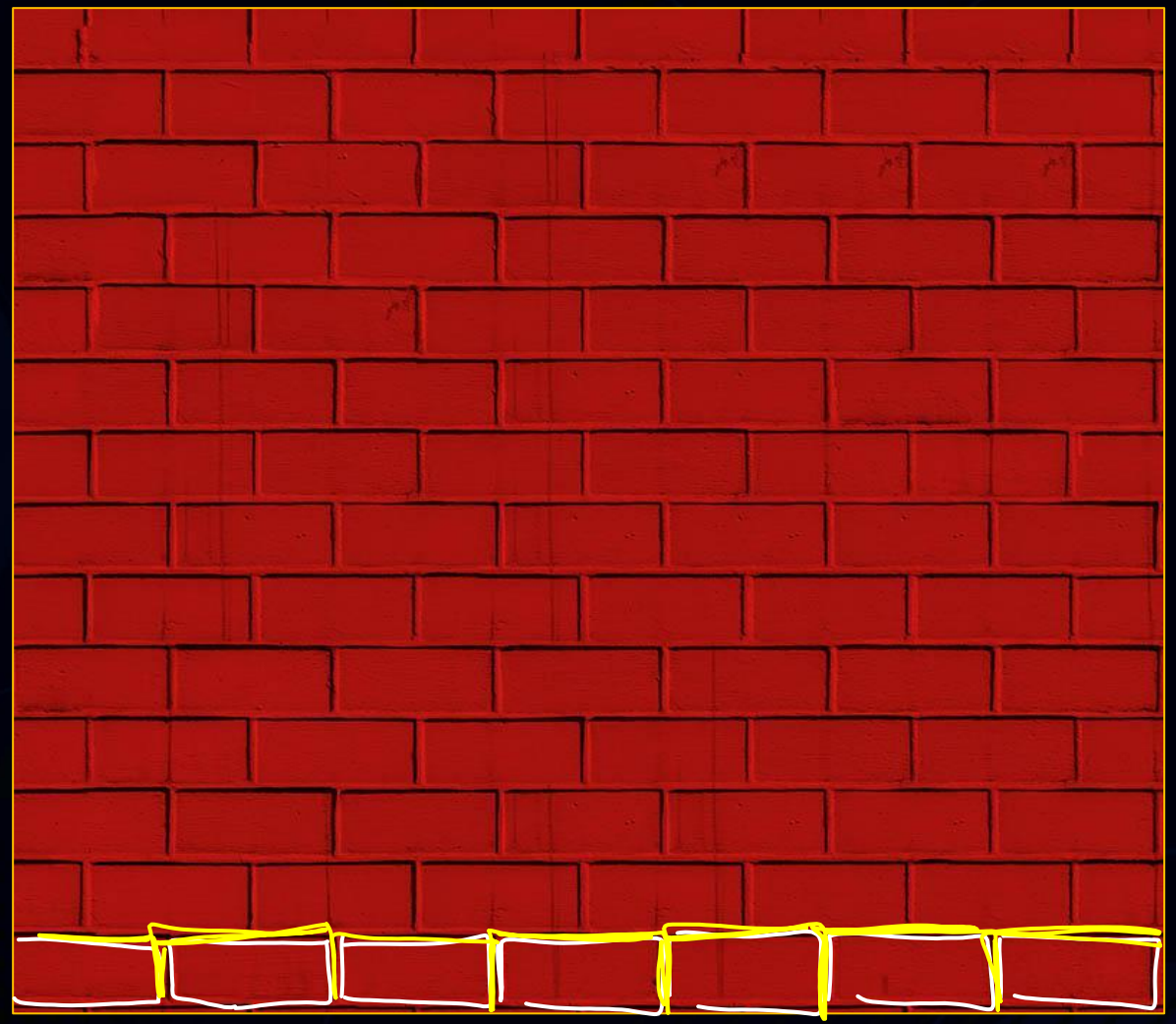
Same type of particles → chemically homogeneous

# Analogy

Red brick: ATOM

Wall: ELEMENT

Cement: FORCE OF ATTRACTION



Element is a type of pure substance which is made from same type of atoms.



# Analogy

Element Types - On basis of no. of atoms

(I) Monoatomic element

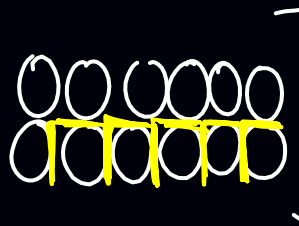
(II) Polyatomic element → atom → two or more than two



Monoatomic element

One

Imp: never exist as molecules

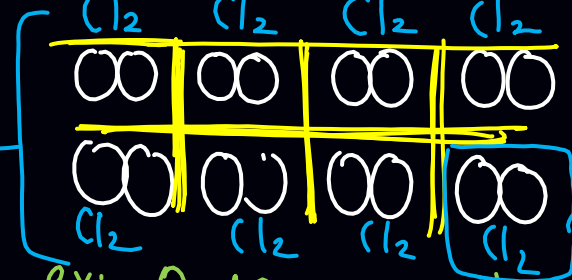


→ Element

ex: Noble gases, Metals etc.



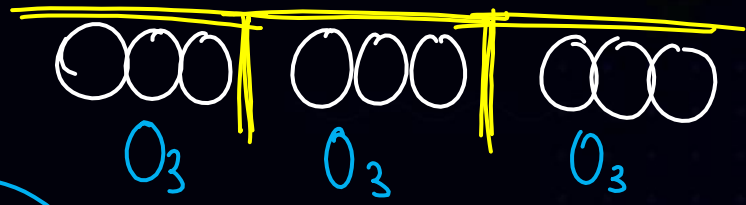
Diatomic element




ex: Oxygen gas, nitrogen gas, hydrogen, chlorine gas. etc.



Triatomic element




ex: Ozone gas

Molecule of element

Molecule: A chemical species that contains two or more than atoms that are chemically bonded to each other.

'same/different'

(*) Independent existence & shows the property of entire element or compound.

# Practice and PYQ's Wallah



PW Ka ChemStar!

## Question

Who coined the term element in 1661?

A John Dalton

B Robert Boyle

C Robert Hooke

D Antoine Laurent Lavoisier

## Question

Which among the following can be defined as the smallest particle of an element that is capable of an independent existence?

*SSC JE 2018*

A. Electron

B. Proton

C. Neutron

D. Molecule

**Question**

Which of the following is a molecule of an element?

RRB JE 2015

A. Neon gas → Ne → Noble gas → Monoatomic element → don't exist as molecule

B. Chlorine gas → $(I_2)$ → molecule of element

C. Sulphur dioxide gas $(SO_2)$ → compound

D. Water liquid $(H_2O)$ → compound

**Question**

Which of the following is a molecule of an element?

*RRB JE 2015*

A $Cl_2$

B $H_2S$

C NaCl

D $CO_2$

not element

# Concept Bites – Ions and Its Types

# Ion

Atom → electrically neutral → no. of protons = no. of electrons

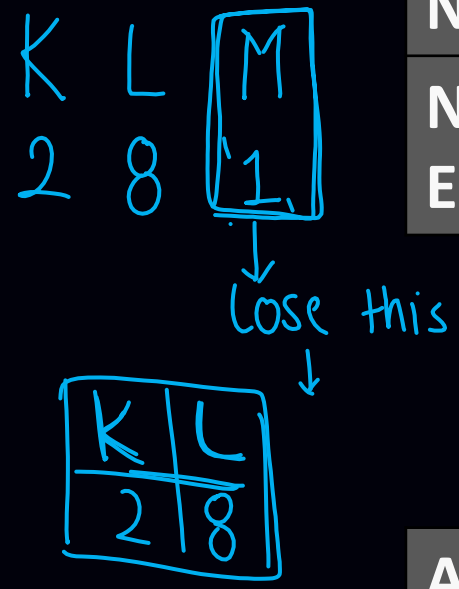
imbalance (बराबर ना हो) → Ion (आयन)

An atom or a group of atoms that contains an overall electrical charge.

- Cation (धनायन)
  - +vely charged
  - No. of protons > electrons
- Anion (ऋणायन)
  - -vely charged
  - No. of protons < electrons

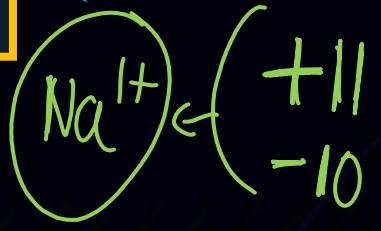
# Example of an Ion

Sodium (Na)




| | |
|---|---|
| Atomic Number | 11 |
| Number of Protons | 11 |
| Number of Electrons | 11 |

Sodium cation ($Na^{1+}$)




| | |
|---|---|
| Atomic Number | 11 |
| Number of Protons | 11 |
| Number of Electrons | 10 |

# Classification of Ions - On the Basis of Number of Atoms

## Monoatomic Ions

One

ex: $Na^{1+}$, $Mg^{2+}$, $Al^{3+}$, $F^{1-}$, $Cl^{1-}$, $O^{2-}$ etc.

⑦ $NO_3^{1-}$ Nitrate ion, ⑧ $NO_2^{1-}$ Nitrite ion, ⑨ $SO_3^{2-}$ Sulphite, ⑩ $HSO_4^{1-}$ Hydrogen sulphate

## Polyatomic Ions

atom, overall charge

Two or more than two

ex: ① $OH^{1-}$ (Hydroxide ion), ② $CO_3^{2-}$ (Carbonate ion), ③ $NH_4^{1+}$ (Ammonium ion)

④ $SO_4^{2-}$ (Sulphate ion), ⑤ $HCO_3^{1-}$ Hydrogen carbonate ion, ⑥ $PO_4^{3-}$ (phosphate ion)

# Practice and PYQ's Wallah



PW Ka ChemStar!

## Question

A cation is formed when _____.

↳ '+vely charged'

Cl

| K | L | M |
|---|---|---|
| 2 | 8 | 7 |

↓ 1 gain

$Cl^-$

| K | L | M |
|---|---|---|
| 2 | 8 | 8 |

no. of protons → 17

– 11 – electrons → 18

A. number of electrons = number of protons

B. number of electrons > number of protons ⟹ Anion

C. number of electrons < number of protons

D. number of neutrons = number of electrons

**Question**

How many electrons are there in $Mg^{2+}$ cation?

A. 9 electrons

B. 10 electrons ✓

C. 11 electrons

D. 12 electrons

K L M

Mg 2 8 2

no. of protons → 12

— " — electrons → 12

$Mg^{2+}$ → 2 protons more than no. of electrons




no. of protons → 12 +12

— " — electrons → 10 -10

2+

# Efficiency Hack!

Tips to Sleep While Studying!

(i) Don’t be too comfortable while studying.

(ii) Read aloud when feeling sleepy.

(iii) Stay hydrated.




#sbsathnai
#pwsathnai
Thank
You !